



The mother of all PC components—the motherboard—is the backbone of your PC. It's also called the mainboard (or 'mobo' for short), and holds the processor, memory and any add-on boards. It's located inside the cabinet and is the component all your peripherals plug into. If you open up the case, it's the thing that looks like a big circuit board—usually green, but it can be any colour.

Oddly enough, the motherboard seldom receives much attention. Early PCs had minimum integrated devices, generally ports for a keyboard and a cassette deck for storage. Everything else, including a display adapter and floppy or hard disk controllers, were add-on components that were connected via expansion slots.

Over time, more devices have been integrated into the motherboard. It's a slow trend though—I/O ports and disk controllers were often mounted on expansion cards as recently as 1995. Other components, such as graphics, networking, SCSI and sound, usually remain discrete.

Several manufacturers have experimented with various levels of integration, building in some or even all these components. However, there are drawbacks. It's hard to upgrade specifications if an integrated component can't

be removed, and highly integrated motherboards often require non-standard cabinets. Moreover, replacing a single faulty component may mean buying a new motherboard.

Consequently, those parts of the system whose specification changes fastest—such as RAM, CPU or graphics—tend to remain in sockets or slots for easy replacement. Similarly, parts that not all users need, such as networking or SCSI, are usually left out of the base specification to keep costs down.

In the late 1990s, the trend was towards putting peripherals designed as integrated chips directly on to the motherboard. Initially, this was confined to audio and graphic chips—obviating the need for separate sound or graphic cards—but in time the peripherals integrated in this way became more diverse, and included items such as SCSI, LAN and even RAID controllers. While there are cost benefits to this approach, the biggest downside is the restriction of future upgrade options.

A board with onboard features is more popular among branded PCs. If you buy a system from a big computer manufacturer, you're likely to get a fairly recent motherboard. You'll find that manufacturers often switch motherboards and other components without notice. They put in the latest technology they can find. However, if you buy your machine from a local dealer, you need to carefully select the right motherboard, as such dealers tend to give you a motherboard that's average at best—after all, they need to keep prices low.

While selecting a motherboard, check out specific sections such as USB ports—the more the merrier. If you have a digital video camera and want to connect it to your PC, make sure you ask about FireWire or USB 2 connectivity. Most motherboards do not come with FireWire ports, even today. FireWire add-on boards are pretty cheap nowadays, so you won't have to pay much extra. Digital video cameras today support USB 2.0, which is much faster than USB 1.x, and comparable to FireWire in speed.

If you're concerned about upgrading, find out how much memory and what type of memory your computer can handle. Ask how many PCI slots the motherboard has, and how many will be filled up when you buy your system.

If you buy a dinky little motherboard that can only take two DDR DIMMs, or have only two USB ports, and you end up needing three, you will be in trouble. So keep this in mind, just in case you need to upgrade suddenly to support new hardware or software. Ideally, you should buy a feature-rich motherboard that doesn't need upgrading often.

What is a chipset?

The word chipset is a term that is commonly used to refer to the core logic of a motherboard. The motherboard is built around a chipset that supports a specific class of processors and a specific type of memory. The main system chipset contains the logic circuits that contain a blueprint of how the processor, memory and other integrated components on the motherboard are meant to communicate. It acts as a hub, controlling the transfer of data between the processor, its cache, system buses and peripherals.

The timely transfer of data without errors is probably the most important function of a system chipset and its proficiency in performing this task dictates exactly how well the system as a whole performs. Apart from the main system chipset which handles the core components (and is commonly referred to as the North Bridge), most motherboards also have a separate South Bridge that integrates, manages and controls all the 'external' peripherals such as IDE, ISA, PCI, USB, etc.

A chipset defines the feature set and the performance of a motherboard—and, by extension, the entire system. The maximum speed of a processor that the chipset supports is important; for any further upgrades you'll need to change only the processor. For example, if a chipset supports up to 2.8 GHz CPUs, and you decide to buy a 2 GHz, you could upgrade to a 2.8 GHz CPU over a period of time without having to invest in a new motherboard.

The type of memory that a chipset supports is also important because there are three types of memory running at different speeds—SDRAM (the slowest type), DDR SDRAM (faster than SDRAM) and RDRAM (theoretically the fastest). The DDR variant is all set to become the new standard for memory as its prices are falling and RDRAM is being increasingly marginalised.